

# FILTER DESIGN TEST CASE

SAW (Surface Acoustic Wave) - Band 71 (600 MHz) for 4G LTE Carrier Aggregation  
Design Validation and Performance Characterization

## DOCUMENT INFORMATION

|                  |                      |
|------------------|----------------------|
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| Author:          | Ana Reyes            |
| Reviewer:        | Jennifer Kim         |
| Design Center:   | Irvine Design Center |
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| Status:          | FAIL                 |
| Priority:        | P2 - High            |

## 1. OBJECTIVE

This document describes the design, simulation, and characterization of a SAW (Surface Acoustic Wave) filter targeting Band 71 (600 MHz) for 4G LTE Carrier Aggregation. The filter is designed on ScAlN on Si substrate with a target center frequency of 3559.2 MHz and bandwidth of 85.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

## 2. SCOPE

- Filter Type: SAW (Surface Acoustic Wave)
- Frequency Band: Band 71 (600 MHz)
- Application: 4G LTE Carrier Aggregation
- Substrate: ScAlN on Si
- Package: QFN (Quad Flat No-Lead)
- Design Tool: PathWave EM Design
- Design Center: Irvine Design Center

This specification applies to all variants of the SAW (Surface Acoustic Wave) design targeting Band 71 (600 MHz). Results are used to qualify the design for mass production release.

## 3. DESIGN PARAMETERS

|                          |                |
|--------------------------|----------------|
| Center Frequency:        | 3559.2 MHz     |
| Bandwidth (3 dB):        | 85.8 MHz       |
| Insertion Loss Target:   | $\leq 1.9$ dB  |
| Return Loss Target:      | $\geq 20.0$ dB |
| Out-of-Band Rejection:   | $\geq 40.0$ dB |
| Isolation (Duplexer):    | $\geq 43.0$ dB |
| Q Factor:                | 4107           |
| Temperature Coefficient: | -11.6 ppm/°C   |
| Die Size:                | 1.8 x 1.7 mm   |
| Wafer Size:              | 8-inch         |
| Number of Resonators:    | 4              |
| Electrode Material:      | Al             |

Passivation: SiO2

4. TEST PROCEDURE

1. Open PathWave EM Design and load the SAW (Surface Acoustic Wave) template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3559.2 MHz, BW 85.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, ScAlN on Si).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10678 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.9 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 129 MHz offset >= 40.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 5.2 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-04-24  
Devices Tested: 27  
Insertion Loss (meas): 1.63 dB  
Return Loss (meas): 22.5 dB  
Rejection (meas): 43.5 dB  
Isolation (meas): 48.3 dB  
Q Factor (meas): 4107  
Group Delay Variation: 4.7 ns  
Power Handling: +30 dBm  
Die Yield: 84.6%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The SAW (Surface Acoustic Wave) design demonstrated below-target performance for Band 71 (600 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

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- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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|-----------|-----------------|-------|------------|
| Author:   | Ana Reyes       | Date: | 2024-04-24 |
| Reviewer: | Jennifer Kim    | Date: | 2024-04-25 |
| Approver: | Dr. Yuko Ishida | Date: | 2024-04-27 |

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